Optimal control

Model answers

Question 1

a) The Lagrangian is $L = (y_1(x))^2 + y_1'(x)y_2'(x) + (y_2'(x))^2$. [3 marks]

b) The Euler-Lagrange equation as one of the first-order necessary conditions for optimality is as follows component-wise:

$$L_{y_1} = \frac{d}{dx} L_{y_1'}$$

and

$$L_{y_2} = \frac{d}{dx} L_{y_2'}$$

which, by substituting L, give respectively

$$2y_1(x) - y_2''(x) = 0 \tag{1}$$

and

$$y_1''(x) + 2y_2''(x) = 0 \tag{2}$$

Next, we consider the first variation of the functional J as given in the notes. Since the final x ($x_f = b = \frac{3\pi}{4}$) is specified, the horizontal perturbation of the terminal point is zero. Furthermore, since $y_1(\frac{3\pi}{4})$ is specified, the vertical perturbation of $y_1(\frac{3\pi}{4})$ is zero. However, $y_2(\frac{3\pi}{4})$ is free, therefore the vertical perturbation of $y_2(\frac{3\pi}{4})$, $\eta_2(\frac{3\pi}{4})$, is arbitrary. Thus $\eta(\frac{3\pi}{4}) = \left(0, \eta_2(\frac{3\pi}{4})\right)^T$. This leads to the additional necessary condition for optimality, $L_{y_2'}\left(\frac{3\pi}{4}, y(\frac{3\pi}{4}), y'(\frac{3\pi}{4})\right) = 0$, which yields

$$y_1'\left(\frac{3\pi}{4}\right) + 2y_2'\left(\frac{3\pi}{4}\right) = 0 \tag{3}$$

[10 marks]

c) By using (1) and (2) gives

$$y_1''(x) + 4y_1(x) = 0$$

which has the solution

$$y_1(x) = A_1 \cos(2x) + B_1 \sin(2x) \tag{4}$$

where $A_1 \in \mathbb{R}$ and $B_1 \in \mathbb{R}$ are two constants of integration. From (1) we obtain

$$y_2'' = 2A_1 \cos(2x) + 2B_1 \sin(2x)$$

which can be integrated twice to yield

$$y_2(x) = -\frac{A_1}{2}\cos(2x) - \frac{B_1}{2}\sin(2x) + C_1 x + D_1 \tag{5}$$

where $C_1 \in \mathbb{R}$ and $D_1 \in \mathbb{R}$ are two further constants of integration. [4 marks]

Answers

d) By using (4) and (5), yields

$$y_1'(x) + 2y_2'(x) = 2C_1$$

therefore from (3), $C_1 = 0$. The specified boundary conditions further yield

$$y_1(0) = A_1 = 2$$

$$y_1\left(\frac{3\pi}{4}\right) = -B_1 = 3$$

$$y_2(0) = -\frac{A_1}{2} + D_1 = -1 + D_1 = 3$$

Summarising, $A_1 = 2$, $B_1 = -3$, $C_1 = 0$, and $D_1 = 4$. Therefore the extremal curve is

$$y_1(x) = 2\cos(2x) - 3\sin(2x)$$

and

$$y_2(x) = -\cos(2x) + \frac{3}{2}\sin(2x) + 4$$

[3 marks]

Answers

Question 2

a) The Hamilton-Jacobi-Bellman (HJB) differential equation for this problem is

$$-V_t(t,x) = \inf_{u\in U}\left\{(x(t)-\cos t)^2 + u(t)^2 + V_x(t,x)u(t)\right\}$$

Since the expression inside the infimum is polynomial in u, we can easily find the control that achieves the infimum: differentiating with respect to u, we have

$$2u(t) + V_x(t,x) = 0$$

which yields

$$u(t) = -\frac{V_x(t,x)}{2} \tag{6}$$

Note that by differentiating the expression inside the infimum once again with respect to u, gives $2 > 0$ and, therefore, the control in (6) does achieve the infimum. Substituting the control (6) into the HJB equation, one obtains

$$-V_t(t,x) = (x(t)-\cos t)^2 - \frac{(V_x(t,x))^2}{4}$$

[5 marks]

b) Based on the terminal cost of $x(t_f)^2$, the boundary condition that must be satisfied by the value function is

$$V(t_f,x) = x^2 \qquad \forall x \in \mathbb{R}$$

[2 marks]

c) i) In order for the assumed solution to match the boundary condition in part b), make

$$K_1(t_f) = 1 \qquad K_2(t_f) = 0 \qquad K_3(t_f) = 0 \tag{7}$$

[2 marks]

ii) By differentiating the assumed solution $V(t,x) = K_1(t)x(t)^2 + 2K_2(t)x(t) + K_3(t)$ one obtains

$$V_x(t,x) = 2K_1(t)x(t) + 2K_2(t) \tag{8}$$

and

$$V_t(t,x) = \dot{K}_1(t)x(t)^2 + 2\dot{K}_2(t)x(t) + \dot{K}_3(t)$$

Then by substituting these terms in the HJB equation, yields

$$\left(\dot{K}_1(t) + 1 - K_1(t)^2\right)x(t)^2 + 2\left(\dot{K}_2(t) - \cos t - K_1(t)K_2(t)\right)x(t) + \dot{K}_3(t) + \cos^2 t - K_2(t)^2 = 0$$

Since this equation must be satisfied for all $x(t)$, its coefficients need to be zero, and therefore it gives the system of ordinary differential equations for $K_1(t)$, $K_2(t)$ and $K_3(t)$

$$\begin{aligned}
\dot{K}_1(t) &= -1 + K_1(t)^2 && (9)\\
\dot{K}_2(t) &= \cos t + K_1(t)K_2(t) && (10)\\
\dot{K}_3(t) &= -\cos^2 t + K_2(t)^2
\end{aligned}$$

[5 marks]

Answers

iii) (9) can easily be solved by inspection as $K_1(t) = 1$, which also satisfies the required boundary condition $K_1(t_f) = 1$. Then, (10) becomes

$$\dot{K}_2(t) = \cos t + K_2(t)$$

which can be solved, after also using $K_2(t_f) = 0$ in (7), to give

$$K_2(t) = -\frac{1}{2}(\sin t_f - \cos t_f)e^{t-t_f} + \frac{1}{2}(\sin t - \cos t) \tag{11}$$

By considering (6), (8) and the state equation, the optimal state trajectory can be found by solving the differential equation

$$\dot{x}^*(t) + x^*(t) = -K_2(t)$$

which yields

$$x^*(t) = Ae^{-t} + \frac{1}{4}(\sin t_f - \cos t_f)e^{t-t_f} + \frac{1}{2}\cos t$$

where A is a constant of integration. By using the boundary condition $x(0) = 0$, the optimal state trajectory is finally given by

$$x^*(t) = -\frac{1}{4}(\sin t_f - \cos t_f)e^{-t-t_f} - \frac{1}{2}e^{-t} + \frac{1}{4}(\sin t_f - \cos t_f)e^{t-t_f} + \frac{1}{2}\cos t \tag{12}$$

[4 marks]

iv) From (6) and (8), the optimal control law is

$$u^*(t) = -x^*(t) - K_2(t)$$

and by substituting from (11) and (12), the optimal control law is

$$u^*(t) = \frac{1}{4}(\sin t_f - \cos t_f)e^{-t-t_f} + \frac{1}{2}e^{-t} + \frac{1}{4}(\sin t_f - \cos t_f)e^{t-t_f} - \frac{1}{2}\sin t$$

[2 marks]

Answers

Question 3

a) The functional to be minimised is

$$J(v) = \int_0^{t_f} 1 \; dt.$$

[2 marks]

b) The Hamiltonian is

$$H(x(t), u(t), p(t)) = p_1(t)x_3(t) + p_2(t)x_4(t) + p_3(t)\frac{cu(t)}{m}\cos v(t) + p_4(t)\frac{cu(t)}{m}\sin v(t) - 1$$

in which it is assumed the costates have been normalised so that $p_0 = -1$.

[2 marks]

c) v must be selected to maximise

$$H(x(t), u(t), p(t)) = p_1(t)x_3(t) + p_2(t)x_4(t) + p_3(t)\frac{cu(t)}{m}\cos v(t) + p_4(t)\frac{cu(t)}{m}\sin v(t) - 1$$

which, since $u(t) > 0 \; \forall t \in [0, t_f]$ and c and m are positive constants, is equivalent to maximising

$$p_3(t)\cos v(t) + p_4(t)\sin v(t) = \sqrt{p_3^2(t) + p_4^2(t)}\sin(v(t) + \beta(t)) \tag{13}$$

in which $\beta(t)$ satisfies

$$\sin\beta(t) = \frac{p_3(t)}{\sqrt{p_3^2(t) + p_4^2(t)}} \quad \text{and} \quad \cos\beta(t) = \frac{p_4(t)}{\sqrt{p_3^2(t) + p_4^2(t)}}$$

The Hamiltonian is therefore maximised when

$$v^*(t) + \beta^*(t) = \frac{\pi}{2}$$

and the optimal control is

$$v^*(t) = -\beta^*(t) + \frac{\pi}{2} \tag{14}$$

By taking the tan on both sides, the optimal control satisfies

$$\tan v^*(t) = \tan\left(-\beta^*(t) + \frac{\pi}{2}\right) = \frac{1}{\tan\beta^*(t)} = \frac{\cos\beta^*(t)}{\sin\beta^*(t)} = \frac{p_4^*(t)}{p_3^*(t)}$$

Note that the method $H_v = 0$ does not work here, since it cannot be guaranteed that $H_{vv} < 0$. Indeed $H_v = 0$ leads directly to $\tan v(t) = p_4(t)/p_3(t)$, which is also satisfied by a control that <u>minimises</u> the Hamiltonian; in that case it can be found from (13) that $v(t) = -\beta(t) - \frac{\pi}{2}$, which is different from (14). [5 marks]

d) The adjoint equations are

$$\begin{aligned}
\dot{p}_1^*(t) &= -H_{x_1}|_* = 0 \\
\dot{p}_2^*(t) &= -H_{x_2}|_* = 0 \\
\dot{p}_3^*(t) &= -H_{x_3}|_* = -p_1(t) \\
\dot{p}_4^*(t) &= -H_{x_4}|_* = -p_2(t)
\end{aligned} \tag{15}$$

[3 marks]

Answers

e) The first two equations in (15) give

$$p_1^*(t) = A \qquad \text{and} \qquad p_2^*(t) = B$$

where A and B are two constants of integration. By substituting these into the last two equations in (15), yields

$$p_3^*(t) = -At + C$$
$$p_4^*(t) = -Bt + D$$

where C and D are two further constants of integration. The optimal control is therefore given by

$$\tan v^*(t) = \frac{p_4^*(t)}{p_3^*(t)} = \frac{-Bt + D}{-At + C}$$

[4 marks]

f) The solutions of the two problems are very similar, with the only differences being that in the present problem the Hamiltonian doesn't include the constant -1, and the boundary conditions for the costates, when solving the adjoint equations, are different. The similarity in the Hamiltonians of the two problems implies that the Hamiltonian maximisation condition is identical for both problems. The integration of the adjoint equations leads to the same solution form of the costates in both problems with the only difference being that the different boundary conditions of the costates lead to different values of the constants of integration, say $\tilde{A}$, $\tilde{B}$, $\tilde{C}$, $\tilde{D}$ instead of A, B, C, D. The optimal solution $v^*(t)$ for the control is therefore given by the same expression as for the previous problem with different values of the constant parameters, as

$$\tan v^*(t) = \frac{p_4^*(t)}{p_3^*(t)} = \frac{-\tilde{B}t + \tilde{D}}{-\tilde{A}t + \tilde{C}}$$

[4 marks]

Answers

Question 4

a) The Hamilton-Jacobi-Bellman (HJB) differential equation for this problem is

$$-V_t(t,x) = \inf_{u\in[0,1]} \Big\{ -(1-u(t))x(t) + V_x(t,x)\alpha u(t)x(t) \Big\}$$

[2 marks]

b) The expression inside the infimum can be rewritten as

$$-x(t) + (1 + V_x(t,x)\alpha)x(t)u(t)$$

therefore the infimum is achieved by setting

$$u(t) = \begin{cases} 1 & \text{for } (1+V_x(t,x)\alpha)x(t) < 0 \\ 0 & \text{for } (1+V_x(t,x)\alpha)x(t) > 0 \\ \tilde{u} & \text{for } (1+V_x(t,x)\alpha)x(t) = 0 \end{cases}$$

where $\tilde{u}$ is an arbitrary value in $[0,1]$, but since $x_0 > 0$ and $\alpha > 0$, it follows that $x(t) > 0$ for all $t \in [0, t_f]$, therefore

$$u(t) = \begin{cases} 1 & \text{for } 1+V_x(t,x)\alpha < 0 \\ 0 & \text{for } 1+V_x(t,x)\alpha > 0 \\ \tilde{u} & \text{for } 1+V_x(t,x)\alpha = 0 \end{cases} \tag{16}$$

[3 marks]

c) By substituting the control (16) into the HJB equation, one obtains

$$\begin{cases} -V_t(t,x) = V_x(t,x)\alpha x(t) & \text{for } 1+V_x(t,x)\alpha < 0 \\ -V_t(t,x) = -x(t) & \text{for } 1+V_x(t,x)\alpha \geq 0 \end{cases} \tag{17}$$

[2 marks]

d) Based on the terminal cost of $-x(t_f)$, the boundary condition that must be satisfied by the value function is

$$V(t_f, x) = -x \qquad \text{for all } x > 0$$

[2 marks]

e) i) In order for the assumed solution to match the boundary condition in part d), make

$$K(t_f) = -1 \tag{18}$$

[2 marks]

ii) By differentiating the assumed solution $V(t,x) = K(t)x(t)$ one obtains

$$V_x(t,x) = K(t) \tag{19}$$

and

$$V_t(t,x) = \dot{K}(t)x(t)$$

Answers

Then by substituting these terms in the HJB equation (17), yields

$$\begin{cases} -\dot{K}(t)x(t) = K(t)\alpha x(t) & \text{for } K(t) < -\frac{1}{\alpha} \\ -\dot{K}(t)x(t) = -x(t) & \text{for } K(t) \geq -\frac{1}{\alpha} \end{cases}$$

Since this equation must be satisfied for all $x(t) > 0$, it gives the differential equation for $K(t)$

$$\begin{cases} -\dot{K}(t) = K(t)\alpha & \text{for } K(t) < -\frac{1}{\alpha} \\ -\dot{K}(t) = -1 & \text{for } K(t) \geq -\frac{1}{\alpha} \end{cases}$$

which can be solved to give

$$K(t) = \begin{cases} Ae^{-\alpha t} & \text{for } K(t) < -\frac{1}{\alpha} \\ t + B & \text{for } K(t) \geq -\frac{1}{\alpha} \end{cases} \tag{20}$$

for some constants A and B. By using the boundary value (18) into (20) (noting that $K(t_f) = -1 > -\frac{1}{\alpha}$ since $0 < \alpha < 1$), gives $-1 = t_f + B$, therefore $B = -1 - t_f$. Therefore, going backwards in time from the final time, $K(t) = t - 1 - t_f$ will decrease from -1 to $-\frac{1}{\alpha}$ at time

$$t_1 = 1 + t_f - \frac{1}{\alpha}$$

At that time there will be a switch to $K(t) = Ae^{-\alpha t}$. Due to the continuity of $K(t)$ at the switching point (C^1 condition), $Ae^{-\alpha t_1} = -\frac{1}{\alpha}$, therefore $A = -\frac{e^{\alpha t_1}}{\alpha}$. Thus

$$K(t) = \begin{cases} -\frac{1}{\alpha}e^{-\alpha(t-t_1)} & \text{for } 0 \leq t < t_1 \\ t - 1 - t_f & \text{for } t_1 \leq t < t_f \end{cases} \tag{21}$$

Note that there will not be any further switches because $-\frac{1}{\alpha}e^{-\alpha(t-t_1)}$ continues to decrease from $-\frac{1}{\alpha}$ with time going backwards from t_1. Note also that $V_x(t,x)$ and $V_t(t,x)$ are continuous at the switching time, thus fulfilling the C^1 condition on $V(t,x)$. [7 marks]

iii) From (16), (19), (20) and (21) the optimal control law is

$$u^*(t) = \begin{cases} 1 & \text{for } 0 \leq t < 1 + t_f - \frac{1}{\alpha} \\ 0 & \text{for } 1 + t_f - \frac{1}{\alpha} < t \leq t_f \end{cases}$$

[2 marks]

Answers